

USMAN ALI SHAHID

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SUMMARY

Computer Science undergraduate at FAST-NUCES (CGPA: 2.95) and AWS Certified Solutions Architect (SAA-C03) specializing in cloud-native infrastructure, Kubernetes orchestration, GitOps automation, and autonomous multi-agent AI systems with Model Context Protocol (MCP). Experienced in building resilient distributed backends, automating CI/CD pipelines, and deploying containerized microservices under strict resource boundaries.

EXPERIENCE

Cloud & DevOps Engineering Intern — Digitalsofts (Pvt.) Ltd.

July 2026 – September 2026 | On-Site

Sitara Techno Park, Faisalabad · Cloud Infrastructure & Platform Engineering

- Architected containerized microservices on single-node Kubernetes (k3s) with Docker under tight memory limits (~3.5 GB RAM), configuring Helm charts, resource quotas, and persistent volumes.
- Implemented declarative GitOps synchronization via ArgoCD and built automated multi-stage GitHub Actions CI/CD workflows for Kustomize manifest validation and container image builds.
- Engineered autonomous multi-agent AI collaboration workflows utilizing LangGraph, CrewAI, and Model Context Protocol (MCP) servers to automate codebase reviews and testing in sandboxes.
- Configured infrastructure telemetry and distributed logging pipelines using Prometheus, Grafana, Loki, and Tempo.

PROJECTS

Self-Hosted AI Platform on Kubernetes — k3s, Helm, Kustomize, ArgoCD, Docker, Vault, KEDA

[Repo ↗](#)

- Architected GitOps AI platform on single-node k3s, deploying 20+ services (PostgreSQL, Redis, Qdrant, MinIO, Vault, n8n) via ArgoCD within an 8GB RAM envelope.
- Implemented full-stack observability (Prometheus/Grafana/Loki/Tempo), cert-manager TLS automation, HashiCorp Vault secrets, and KEDA autoscaling.

AI Ops Console & Microservice Backend — FastAPI, Docker, PostgreSQL, Redis, CI/CD, ARQ

[Repo ↗](#)

- Built async FastAPI backend with JWT/RBAC security, Alembic PostgreSQL, Redis caching/rate limiting, and ARQ background workers containerized via Docker Compose.
- Configured automated GitHub Actions testing and container publishing pipelines with Prometheus/Grafana metric instrumentation.

Smart City Traffic & Transport AI Systems — Python, Flask, Oracle 11g, BFS/A* Search, Matplotlib

[Repo ↗](#)

- Engineered multi-agent traffic grid simulation routing emergency vehicles using heuristic search (BFS, UCS, A*) alongside an Oracle 11g relational transit database.
- Built real-time visualization measuring algorithm performance, node expansions, path latency, and commuter transit scheduling queries.

Atari Breakout Arcade Engine — 16-bit x86 Assembly (8086), Direct Video Memory, BIOS Interrupts

[Repo ↗](#)

- Engineered classic arcade game from scratch in 16-bit x86 Assembly manipulating Mode 13h video RAM directly for high-speed frame rendering and collision physics.
- Configured custom BIOS keyboard interrupt handlers (INT 16h / INT 09h) ensuring responsive real-time hardware-level controls.

Autonomous Wumpus World AI Agent — Python, Propositional Logic, Knowledge-Based Reasoning, Search

[Repo ↗](#)

- Developed an autonomous knowledge-based agent in Python utilizing propositional logic theorem proving and resolution to safely infer world state under partial observability.
- Integrated heuristic pathfinding enabling the agent to evade lethal hazards and navigate optimal gold retrieval paths with zero human intervention.

Additional Projects: [Repo Scanner I](#) (Python DevSecOps CLI) · [CDAF](#) (Discrete Math Framework in C++) · [SearchScope](#) (Algorithm Visualizer)

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA-C03) — Amazon Web Services

Issued: August 2026 | Score: 790 / 1000 (Pass)

- Validated competency in designing resilient multi-tier cloud architectures, high-availability VPC networks, secure compute & storage (EKS, EC2, S3, RDS), disaster recovery, and cost optimization.

SKILLS

Languages:	Python, C/C++, SQL, JavaScript, HTML5/CSS3, Bash, x86 Assembly
Cloud & DevOps:	AWS, Kubernetes (k3s), Docker & Compose, Helm, Kustomize, ArgoCD (GitOps), Jenkins, CI/CD, Terraform IaC, Linux
AI & Backend:	Model Context Protocol (MCP), LangGraph, CrewAI, FastAPI, Flask, Redis, Qdrant Vector DB, n8n, React, NumPy
Observability & Tools:	Prometheus, Grafana, Loki, Tempo, OpenTelemetry, HashiCorp Vault, cert-manager, KEDA, Git, GitHub, VS Code

EDUCATION

BS in Computer Science — FAST – National University of Computer & Emerging Sciences (NUCES)

September 2024 – June 2028

CGPA: 2.95 / 4.0 | Relevant Coursework: Cloud Computing, Distributed Systems, Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks